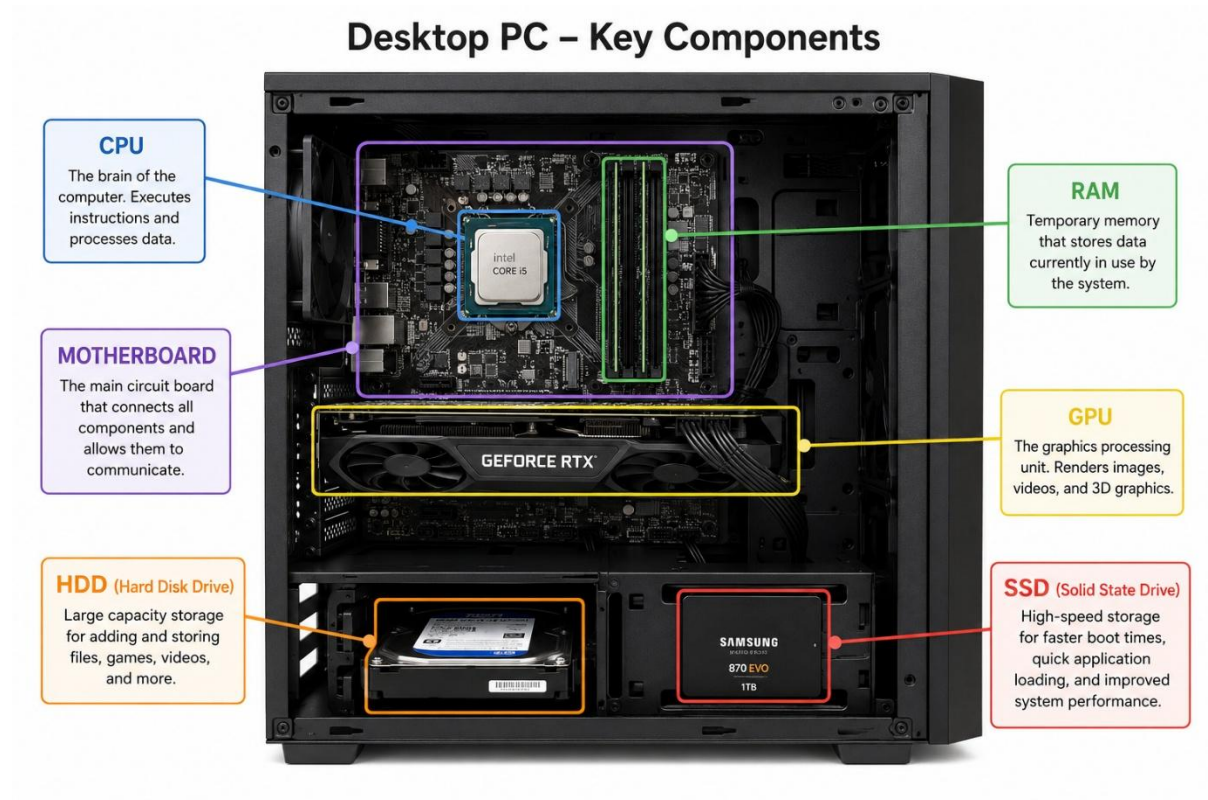


1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	SSD (Solid-State Drive)	HDD (Hard Disk Drive)
Random Access Time	0.1 ms	5–10 ms
Read Latency Time	Very Low	High
Data Transfer Speed	100 MB/s to 500 MB/s	50 MB/s to 100 MB/s
Reliability	High Reliability	Low Reliability
Moving Parts	No moving parts to fail	Has moving parts and is subject to sudden failure
Size & Weight	Small and Lightweight	Relatively Large and Heavy
Storage Capacity	Available up to 2 TB	Available up to 4 TB
Power Consumption	2 Watts	6–12 Watts
Typical Use Cases	Operating Systems, Applications, Gaming, Laptops	Data Storage, Backups, Media Files, Archives

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit)

The CPU is the main processor of the computer that executes game instructions and calculations. In games like PUBG or Free Fire, it handles tasks such as player movements, game logic, AI behavior, and communication between different hardware components.

RAM (Random Access Memory)

RAM temporarily stores the game data and files that are currently in use. More RAM allows PUBG or Free Fire to load maps, textures, and game assets quickly, resulting in smoother gameplay and faster loading times.

GPU (Graphics Processing Unit)

The GPU is responsible for rendering graphics, images, and visual effects in the game. In PUBG or Free Fire, it processes high-quality textures, lighting, shadows, and animations to provide smooth and realistic visuals.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Hardware Component	Why Upgrade It?
CPU (Processor)	A faster CPU improves game performance, handles game logic, AI, and reduces lag during gameplay.
GPU (Graphics Processing Unit)	A better GPU provides higher frame rates, improved graphics quality, and smoother gameplay.
RAM (Memory)	Increasing RAM helps the game load faster and allows smoother multitasking while gaming.
SSD (Solid-State Drive)	An SSD reduces loading times, speeds up game installation, and improves overall system responsiveness.
Power Supply Unit (PSU)	A higher-capacity PSU may be needed to provide enough power for upgraded components, especially a new GPU.
Cooling System	Better cooling prevents overheating and helps maintain stable performance during long gaming sessions.
Monitor (Optional)	A monitor with a higher refresh rate and resolution can provide a smoother and more enjoyable gaming experience.